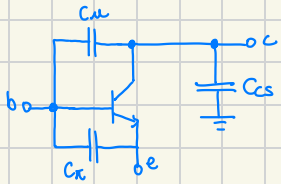
ELG 3136

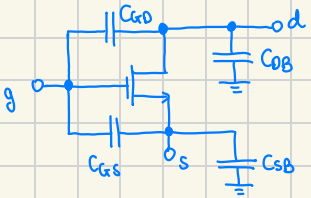
Lecture 5

Intrinsic Capacitance of BJT and MOSFET

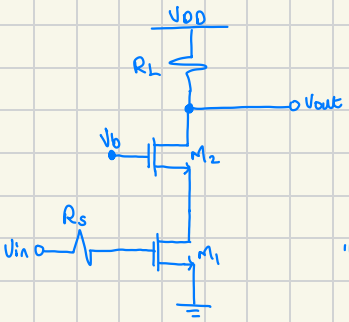
BJT



MOSFET

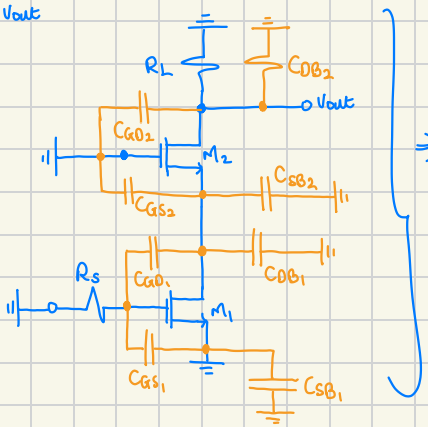


Poles of MOS Cascode

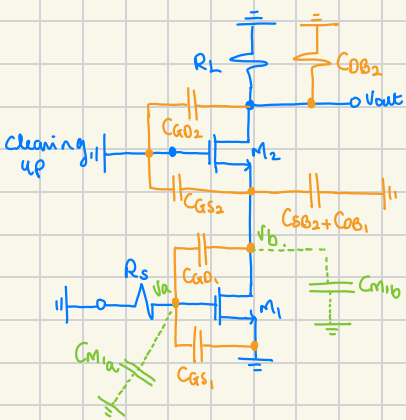


STEP 1: Disable all DC and V_{in}

STEP 2: Draw intrinsic capacitances of transistors

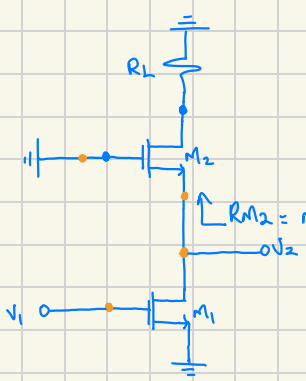


⇒ Cleaning up

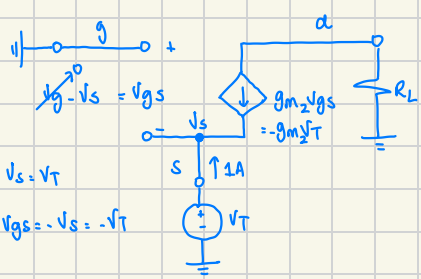


$C_{M1a} = (1 - A_{V_1}) C_{GD_1}$, $C_{M1b} = \left(1 - \frac{1}{A_{V_1}}\right) C_{GD_1}$, $A_V = \frac{V_b}{V_a}$

Finding Rout



Using classic method



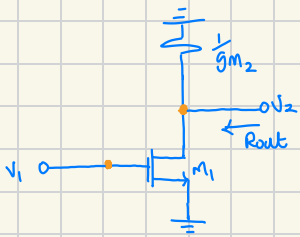
$V_s = V_T$

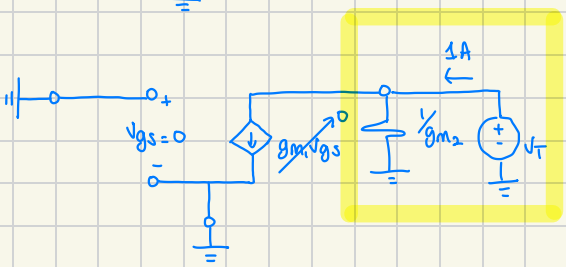
$\Rightarrow V_{gs} = -V_s = -V_T$

$g_{m_2} V_{gs} = -1 \Rightarrow +g_{m_2} V_T = +1$

$\Rightarrow \boxed{V_T = \frac{1}{g_{m_2}}}$

$\Rightarrow \boxed{R_{M_2} = \frac{1}{g_{m_2}}}$





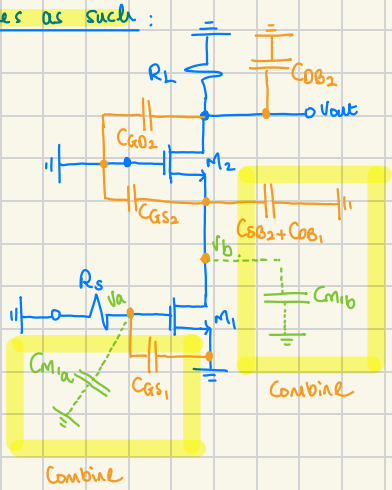
$V_T = \frac{1}{g_{m2}} \rightarrow \boxed{R_{out} = \frac{1}{g_{m2}}}$

$\rightarrow A_{v_1} = -g_{m_1}\left(\frac{1}{g_{m2}}\right) \rightarrow \boxed{A_{v_1} = -\frac{g_{m1}}{g_{m2}}}$

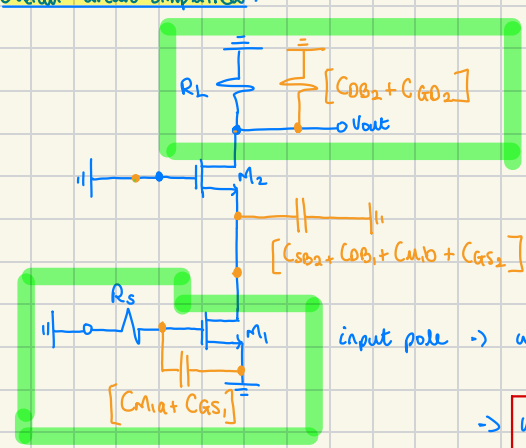
$C_{M1a} = \left(1 + \frac{g_{m1}}{g_{m2}}\right) C_{GD_1} \rightarrow \boxed{C_{M1a} = \left(1 + \frac{g_{m1}}{g_{m2}}\right) C_{GD_1}}$

$C_{M1b} = \left(1 + \frac{g_{m2}}{g_{m1}}\right) C_{GD_2} \rightarrow \boxed{C_{M1b} = \left(1 + \frac{g_{m2}}{g_{m1}}\right) C_{GD_1}}$

Overall circuit becomes as such :



Overall circuit simplified :

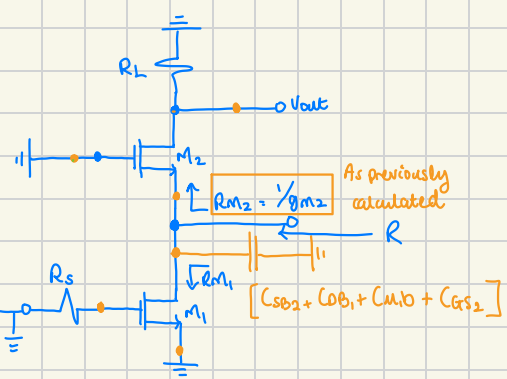


Output pole $\rightarrow \boxed{\omega_{p,out} = \frac{1}{R_L(C_{DB_2} + C_{GD_2})}}$

$[C_{SB_2} + C_{DB_1} + C_{M1b} + C_{GS_2}]$ -> parallel to what ?

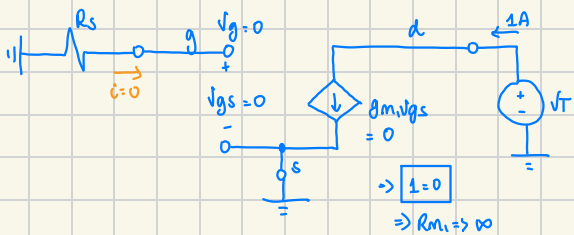
input pole $\rightarrow \omega_{p,in} = \frac{1}{R_S(G_{GS_1} + C_{M1a})} = \frac{1}{R_S\left(C_{GS_1} + \left(1 + \frac{g_{m1}}{g_{m2}}\right)C_{GD_1}\right)}$

$\rightarrow \boxed{\omega_{p,in} = \frac{1}{R_S\left(C_{GS_1} + \left(1 + \frac{g_{m1}}{g_{m2}}\right)C_{GD_1}\right)}}$



R_{M_1} is not readily available in circuit sheet.

Using classic method



$$R = R_{M_2} // R_{M_1} = \frac{1}{\left(\frac{1}{R_{M_1}} + \frac{1}{R_{M_2}}\right)} = \frac{1}{\left(\frac{1}{R_{M_2}}\right)} = R_{M_2} = \frac{1}{g_{m_2}} \Rightarrow \boxed{R = \frac{1}{g_{m_2}}}$$

$$\omega_p = \frac{1}{\left(\frac{1}{g_{m_2}}\right)\left[C_{DB_1} + C_{GS_2} + C_{M_1b} + C_{SB_2}\right]} \Rightarrow \boxed{\omega_p = \frac{1}{\left(\frac{1}{g_{m_2}}\right)\left[C_{DB_1} + C_{GS_2} + \left(1 + \frac{g_{m_2}}{g_{m_1}}\right) C_{GD_1} + C_{SB_2}\right]}}$$

General equation for transfer function :

$$H(\omega) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)\left(1 + \frac{s}{\omega_{p3}}\right) \dots \left(1 + \frac{s}{\omega_{pn}}\right)}$$

A_0 : regular gain of circuit with no capacitors.

ω_{pi} : computed poles.